

MallnSight

Static Malware Analysis & Threat Intelligence Platform

Source Code Listing

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Repository: <https://github.com/vibushasatheeshkumar/mallnsight>

Backend

app.py

```
import os
import uuid

from dotenv import load_dotenv

load_dotenv()

from flask import Flask, render_template, request, send_file, abort
from werkzeug.utils import secure_filename

from analysis.hash import calculate_hashes
from analysis.metadata import get_metadata
from analysis.pe_analysis import analyze_pe
from analysis.entropy import analyze_entropy
from analysis.strings import extract_strings
from analysis.yara_scan import scan_file
from analysis.scoring import calculate_score
from analysis.report import generate_report
from analysis.history import save_analysis, get_recent_analyses, is_available, get_connection_error

app = Flask(__name__)

UPLOAD_FOLDER = "uploads"
REPORTS_FOLDER = "reports"

ALLOWED_EXTENSIONS = {
    ".exe", ".dll", ".sys", ".msi", ".zip", ".apk", ".pdf", ".docx", ".bin"
}

app.config["UPLOAD_FOLDER"] = UPLOAD_FOLDER
app.config["MAX_CONTENT_LENGTH"] = 100 * 1024 * 1024 # 100 MB

# Create uploads/reports folders if they don't exist
os.makedirs(UPLOAD_FOLDER, exist_ok=True)
os.makedirs(REPORTS_FOLDER, exist_ok=True)

# In-memory map of generated PDF reports, keyed by a one-time download id.
_generated_reports = {}
```

```
def is_allowed_file(filename):  
    extension = os.path.splitext(filename)[1].lower()  
    return extension in ALLOWED_EXTENSIONS
```

```
@app.route("/")  
def home():  
    return render_template("home.html")
```

```
@app.route("/about")  
def about():  
    return render_template("about.html")
```

```
@app.route("/features")  
def features():  
    return render_template("features.html")
```

```
@app.route("/upload")  
def upload():  
    return render_template("upload.html")
```

```
@app.route("/dashboard")  
def dashboard():  
    return render_template("dashboard.html")
```

```
@app.route("/contact")  
def contact():  
    return render_template("contact.html")
```

```
@app.route("/analyze", methods=["POST"])  
def analyze():
```

```
    if "file" not in request.files:  
        return "No file uploaded.", 400
```

```
    uploaded_file = request.files["file"]
```

```
    if uploaded_file.filename == "":  
        return "No file selected.", 400
```

```

original_filename = secure_filename(uploaded_file.filename)

if not original_filename:
    return "Invalid filename.", 400

if not is_allowed_file(original_filename):
    return "File type is not supported.", 400

extension = os.path.splitext(original_filename)[1].lower()
stored_filename = f"{uuid.uuid4().hex}{extension}"
file_path = os.path.join(app.config["UPLOAD_FOLDER"], stored_filename)

uploaded_file.save(file_path)

# File Information
metadata = get_metadata(file_path)
metadata["name"] = original_filename

# File Hashes
hashes = calculate_hashes(file_path)

# PE Analysis
pe_info = analyze_pe(file_path)

# Entropy Analysis
entropy_info = analyze_entropy(file_path)

# Suspicious Strings
strings_info = extract_strings(file_path)

# YARA Scan
yara_info = scan_file(file_path)

# Risk Scoring
score_info = calculate_score(
    entropy_info=entropy_info,
    yara_info=yara_info,
    strings_info=strings_info,
    pe_info=pe_info
)

report_path = generate_report(
    metadata, hashes, pe_info, entropy_info, strings_info, yara_info, score_info
)

```

```

download_id = uuid.uuid4().hex
_generated_reports[download_id] = report_path

# Cloud history (MongoDB Atlas) — non-fatal if not configured
save_analysis(metadata, hashes, score_info)

return render_template(
    "dashboard.html",
    metadata=metadata,
    hashes=hashes,
    pe_info=pe_info,
    entropy_info=entropy_info,
    strings_info=strings_info,
    yara_info=yara_info,
    score_info=score_info,
    download_id=download_id
)

@app.route("/history")
def history():
    return render_template(
        "history.html",
        available=is_available(),
        error=get_connection_error(),
        records=get_recent_analyses()
    )

@app.route("/download/<download_id>")
def download_report(download_id):
    report_path = _generated_reports.get(download_id)

    if not report_path or not os.path.exists(report_path):
        abort(404)

    return send_file(report_path, as_attachment=True, download_name="MallnSight_Report.pdf")

if __name__ == "__main__":
    debug_mode = os.environ.get("FLASK_DEBUG", "0") == "1"
    app.run(debug=debug_mode)

```

Analysis Modules

analysis/hash.py

```
import hashlib

def calculate_hashes(file_path):
    """
    Calculate MD5, SHA1 and SHA256 hashes of a file.
    """

    md5 = hashlib.md5()
    sha1 = hashlib.sha1()
    sha256 = hashlib.sha256()

    with open(file_path, "rb") as f:

        while True:

            chunk = f.read(4096)

            if not chunk:
                break

            md5.update(chunk)
            sha1.update(chunk)
            sha256.update(chunk)

    return {
        "md5": md5.hexdigest(),
        "sha1": sha1.hexdigest(),
        "sha256": sha256.hexdigest()
    }
```

analysis/metadata.py

```
import os
import mimetypes

def get_metadata(file_path):

    file_size = os.path.getsize(file_path)

    file_name = os.path.basename(file_path)
```

```
mime_type = mimetypes.guess_type(file_path)[0]

extension = os.path.splitext(file_name)[1]

return {
    "name": file_name,
    "size": round(file_size / 1024, 2),
    "extension": extension,
    "mime": mime_type
}
```

analysis/pe_analysis.py

```
import pefile
import os
from datetime import datetime

def analyze_pe(file_path):
    try:
        pe = pefile.PE(file_path)

        info = {}

        # Architecture
        machine = pe.FILE_HEADER.Machine

        architectures = {
            0x14c: "x86 (32-bit)",
            0x8664: "x64 (64-bit)",
            0x1c0: "ARM",
            0xaa64: "ARM64"
        }

        info["architecture"] = architectures.get(
            machine,
            hex(machine)
        )

        # Compile Time
        info["compile_time"] = datetime.utcfromtimestamp(
            pe.FILE_HEADER.TimeDateStamp
        ).strftime("%Y-%m-%d %H:%M:%S UTC")

        # Entry Point
```

```

info["entry_point"] = hex(
    pe.OPTIONAL_HEADER.AddressOfEntryPoint
)

# Image Base
info["image_base"] = hex(
    pe.OPTIONAL_HEADER.ImageBase
)

# Number of Sections
info["number_of_sections"] = pe.FILE_HEADER.NumberOfSections

# Section Names
sections = []

for section in pe.sections:
    sections.append({
        "name": section.Name.decode(errors="ignore").strip("\x00"),
        "virtual_size": section.Misc_VirtualSize,
        "raw_size": section.SizeOfRawData,
        "entropy": round(section.get_entropy(), 2)
    })

info["sections"] = sections

# Imports
imports = []

if hasattr(pe, "DIRECTORY_ENTRY_IMPORT"):
    for entry in pe.DIRECTORY_ENTRY_IMPORT:
        dll = entry.dll.decode(errors="ignore")

        for imp in entry.imports:
            imports.append({
                "dll": dll,
                "function": (
                    imp.name.decode(errors="ignore")
                    if imp.name
                    else f"Ordinal {imp.ordinal}"
                )
            })

info["imports"] = imports

# Exports

```



```

exports = []

if hasattr(pe, "DIRECTORY_ENTRY_EXPORT"):
    for exp in pe.DIRECTORY_ENTRY_EXPORT.symbols:
        exports.append(
            exp.name.decode(errors="ignore")
            if exp.name
            else f"Ordinal {exp.ordinal}"
        )

info["exports"] = exports

return info

except Exception as e:
    return {
        "error": str(e)
    }

```

analysis/entropy.py

```

import math
from collections import Counter

def calculate_entropy(data):
    """
    Shannon entropy of a byte string, in bits per byte (0.0 - 8.0).
    """

    if not data:
        return 0.0

    counts = Counter(data)
    length = len(data)

    entropy = 0.0

    for count in counts.values():
        p = count / length
        entropy -= p * math.log2(p)

    return entropy

```

```

def analyze_entropy(file_path):
    """
    Compute overall file entropy and classify how likely the file is
    to be packed/encrypted/compressed.
    """

    with open(file_path, "rb") as f:
        data = f.read()

    entropy = round(calculate_entropy(data), 2)

    if entropy >= 7.5:
        verdict = "Very High (likely packed/encrypted)"
    elif entropy >= 6.5:
        verdict = "High (possibly compressed/packed)"
    elif entropy >= 4.0:
        verdict = "Normal"
    else:
        verdict = "Low (mostly repetitive/plain data)"

    return {
        "entropy": entropy,
        "max_entropy": 8.0,
        "verdict": verdict
    }

```

analysis/strings.py

```

import re

MIN_LENGTH = 4
MAX_STRINGS = 500

_ASCII_RE = re.compile(rb"[\x20-\x7e]{%d}" % MIN_LENGTH)
_WIDE_RE = re.compile(rb"([\x20-\x7e]|[\x00]){%d}" % MIN_LENGTH)

SUSPICIOUS_PATTERNS = {
    "Process Injection API": re.compile(
        rb"(CreateRemoteThread|WriteProcessMemory|VirtualAllocEx|"
        rb"NtUnmapViewOfSection|QueueUserAPC|SetThreadContext)\b"
    ),
    "Dynamic Loading API": re.compile(
        rb"(LoadLibraryA|LoadLibraryW|GetProcAddress|LdrLoadDll)\b"
    ),
    "Execution API": re.compile(

```

```

        r"\b(WinExec|ShellExecuteA|ShellExecuteW|CreateProcessA|CreateProcessW)\b"
    ),
    "Memory API": re.compile(r"\b(VirtualAlloc|VirtualProtect|HeapCreate)\b"),
    "Command Interpreter": re.compile(
        r"\b(cmd\.exe|powershell\.exe|wscript\.exe|cscript\.exe|rundll32\.exe|mshta\.exe)\b",
        re.IGNORECASE,
    ),
    "Persistence / Registry": re.compile(
        r"(HKEY_[A-Z_]+\|\\CurrentVersion\\Run|SOFTWARE\\Microsoft\\Windows\\CurrentVersion\\Run)"
    ),
    "URL": re.compile(r"\bhttps?:/[^\s\"'<>]+", re.IGNORECASE),
    "IP Address": re.compile(r"\b(?:\d{1,3}\.){3}\d{1,3}\b"),
    "Anti-Debug / Anti-VM": re.compile(
        r"\b(IsDebuggerPresent|CheckRemoteDebuggerPresent|vmware|virtualbox|sandboxie)\b",
        re.IGNORECASE,
    ),
}

```

```

def extract_strings(file_path, min_length=MIN_LENGTH, max_strings=MAX_STRINGS):

```

```

    """

```

```

    Extract printable ASCII and UTF-16LE strings from a file and flag
    any that match known-suspicious patterns.

```

```

    """

```

```

    with open(file_path, "rb") as f:

```

```

        data = f.read()

```

```

    found = []

```

```

    for match in _ASCII_RE.finditer(data):

```

```

        found.append(match.group().decode("ascii", errors="ignore"))

```

```

    for match in _WIDE_RE.finditer(data):

```

```

        found.append(match.group().decode("utf-16le", errors="ignore"))

```

```

    unique_strings = list(dict.fromkeys(found))

```

```

    suspicious = []

```

```

    for category, pattern in SUSPICIOUS_PATTERNS.items():

```

```

        for s in unique_strings:

```

```

            m = pattern.search(s)

```

```

            if m:

```

```

                suspicious.append({

```

```

        "category": category,
        "value": m.group()
    })

unique_suspicious = list(
    {(s["category"], s["value"]): s for s in suspicious}.values()
)

return {
    "total_strings": len(unique_strings),
    "strings": unique_strings[:max_strings],
    "suspicious": unique_suspicious
}

```

analysis/yara_scan.py

```

import glob
import os

try:
    import yara
    YARA_AVAILABLE = True
except ImportError:
    YARA_AVAILABLE = False

RULES_DIR = os.path.join(os.path.dirname(__file__), "..", "yara_rules")

_compiled_rules = None
_compile_error = None

def _load_rules():
    global _compiled_rules, _compile_error

    if _compiled_rules is not None or _compile_error is not None:
        return

    rule_files = glob.glob(os.path.join(RULES_DIR, "*.yar"))

    if not rule_files:
        _compile_error = "No YARA rule files found."
        return

    try:
        filepaths = {

```

```

        f"rule_{i}": path for i, path in enumerate(rule_files)
    }
    _compiled_rules = yara.compile(filepaths=filepaths)
except yara.Error as e:
    _compile_error = str(e)

def scan_file(file_path):
    """
    Scan a file against the bundled YARA rules.
    Returns a list of {rule, description, severity} matches.
    """

    if not YARA_AVAILABLE:
        return {
            "available": False,
            "error": "yara-python is not installed.",
            "matches": []
        }

    _load_rules()

    if _compile_error:
        return {
            "available": False,
            "error": _compile_error,
            "matches": []
        }

    try:
        raw_matches = _compiled_rules.match(file_path)
    except yara.Error as e:
        return {
            "available": False,
            "error": str(e),
            "matches": []
        }

    matches = []

    for m in raw_matches:
        matches.append({
            "rule": m.rule,
            "description": m.meta.get("description", ""),
            "severity": m.meta.get("severity", "LOW")
        })

```

```

    })

    return {
        "available": True,
        "error": None,
        "matches": matches
    }

```

analysis/scoring.py

```

SEVERITY_WEIGHTS = {
    "HIGH": 25,
    "MEDIUM": 15,
    "LOW": 5
}

def calculate_score(entropy_info=None, yara_info=None, strings_info=None, pe_info=None):
    """
    Combine entropy, YARA matches and suspicious strings/imports into a
    single 0-100 risk score with a human-readable verdict.
    """

    score = 0
    reasons = []

    if entropy_info:
        entropy = entropy_info.get("entropy", 0)

        if entropy >= 7.5:
            score += 30
            reasons.append(f"Very high overall entropy ({entropy})")
        elif entropy >= 6.5:
            score += 15
            reasons.append(f"Elevated overall entropy ({entropy})")

    if yara_info and yara_info.get("matches"):
        for match in yara_info["matches"]:
            weight = SEVERITY_WEIGHTS.get(match.get("severity", "LOW"), 5)
            score += weight
            reasons.append(f"YARA rule matched: {match['rule']}")

    if strings_info and strings_info.get("suspicious"):
        categories = {s["category"] for s in strings_info["suspicious"]}

```

```

    for category in categories:
        score += 8
        reasons.append(f"Suspicious indicator found: {category}")

if pe_info and not pe_info.get("error"):
    for section in pe_info.get("sections", []):
        if section.get("entropy", 0) >= 7.5:
            score += 10
            reasons.append(f"High entropy PE section: {section['name']}")
            break

score = min(score, 100)

if score >= 70:
    verdict = "HIGH RISK"
elif score >= 40:
    verdict = "SUSPICIOUS"
elif score >= 15:
    verdict = "LOW RISK"
else:
    verdict = "CLEAN"

return {
    "score": score,
    "verdict": verdict,
    "reasons": reasons
}

```

analysis/report.py

```

import os
from datetime import datetime, timezone

from reportlab.lib import colors
from reportlab.lib.pagesizes import A4
from reportlab.lib.styles import getSampleStyleSheet
from reportlab.lib.units import cm
from reportlab.platypus import (
    SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle
)

REPORTS_DIR = os.path.join(os.path.dirname(__file__), "..", "reports")

def _table(rows, col_widths=None):

```

```

table = Table(rows, colWidths=col_widths)
table.setStyle(TableStyle([
    ("BACKGROUND", (0, 0), (-1, 0), colors.HexColor("#1C2128")),
    ("TEXTCOLOR", (0, 0), (-1, 0), colors.white),
    ("FONTNAME", (0, 0), (-1, 0), "Helvetica-Bold"),
    ("GRID", (0, 0), (-1, -1), 0.5, colors.HexColor("#30363D")),
    ("FONTSIZE", (0, 0), (-1, -1), 9),
    ("ROWBACKGROUNDS", (0, 1), (-1, -1), [colors.white, colors.HexColor("#F4F4F6")]),
]))
return table

```

```

def generate_report(metadata, hashes, pe_info, entropy_info, strings_info, yara_info, score_info):
    """
    Render the analysis results into a PDF report and return its file path.
    """

```

```

    os.makedirs(REPORTS_DIR, exist_ok=True)

```

```

    base_name = os.path.splitext(metadata.get("name", "report"))[0]
    timestamp = datetime.now(timezone.utc).strftime("%Y%m%d%H%M%S")
    out_path = os.path.join(REPORTS_DIR, f"{base_name}_{timestamp}.pdf")

```

```

    styles = getSampleStyleSheet()
    story = []

```

```

    story.append(Paragraph("MallnSight Static Analysis Report", styles["Title"]))
    story.append(Paragraph(
        datetime.now(timezone.utc).strftime("Generated %Y-%m-%d %H:%M UTC"),
        styles["Normal"]
    ))
    story.append(Spacer(1, 0.5 * cm))

```

```

    story.append(Paragraph(
        f"Verdict: {score_info['verdict']} (Risk score: {score_info['score']}/100)",
        styles["Heading2"]
    ))

```

```

    if score_info.get("reasons"):
        for reason in score_info["reasons"]:
            story.append(Paragraph(f"- {reason}", styles["Normal"]))

```

```

    story.append(Spacer(1, 0.5 * cm))

```

```

    story.append(Paragraph("File Information", styles["Heading2"]))

```



```

story.append(_table([
    ["Property", "Value"],
    ["Filename", metadata.get("name", "")],
    ["Size (KB)", str(metadata.get("size", ""))],
    ["Extension", metadata.get("extension", "")],
    ["MIME Type", str(metadata.get("mime", ""))],
]))
story.append(Spacer(1, 0.5 * cm))

story.append(Paragraph("Hashes", styles["Heading2"]))
story.append(_table([
    ["Type", "Value"],
    ["MD5", hashes.get("md5", "")],
    ["SHA1", hashes.get("sha1", "")],
    ["SHA256", hashes.get("sha256", "")],
], col_widths=[3 * cm, 12 * cm]))
story.append(Spacer(1, 0.5 * cm))

story.append(Paragraph("Entropy", styles["Heading2"]))
story.append(Paragraph(
    f"Overall file entropy: {entropy_info.get('entropy')} / 8.0 "
    f"({entropy_info.get('verdict')})",
    styles["Normal"]
))
story.append(Spacer(1, 0.5 * cm))

if not pe_info.get("error"):
    story.append(Paragraph("PE Analysis", styles["Heading2"]))
    story.append(_table([
        ["Property", "Value"],
        ["Architecture", pe_info.get("architecture", "")],
        ["Compile Time", pe_info.get("compile_time", "")],
        ["Entry Point", pe_info.get("entry_point", "")],
        ["Image Base", pe_info.get("image_base", "")],
        ["Number of Sections", str(pe_info.get("number_of_sections", ""))],
    ]))
    story.append(Spacer(1, 0.5 * cm))

if yara_info.get("matches"):
    story.append(Paragraph("YARA Matches", styles["Heading2"]))
    rows = [
        ["Rule", "Severity", "Description"]
    ]
    for m in yara_info["matches"]:
        rows.append([m["rule"], m["severity"], m["description"]])
    story.append(_table(rows, col_widths=[5 * cm, 3 * cm, 7 * cm]))
    story.append(Spacer(1, 0.5 * cm))

```

```

if strings_info.get("suspicious"):
    story.append(Paragraph("Suspicious Strings", styles["Heading2"]))
    rows = [{"Category", "Value"}]
    for s in strings_info["suspicious"]:
        rows.append([s["category"], s["value"]])
    story.append(_table(rows, col_widths=[6 * cm, 9 * cm]))

doc = SimpleDocTemplate(out_path, pagesize=A4)
doc.build(story)

return out_path

```

analysis/history.py

```

import os
from datetime import datetime, timezone

try:
    from pymongo import MongoClient
    from pymongo.errors import PyMongoError
    PYMONGO_AVAILABLE = True
except ImportError:
    PYMONGO_AVAILABLE = False

_client = None
_collection = None
_connection_error = None
_attempted = False

def _get_collection():
    """
    Lazily connect to MongoDB Atlas on first use (env vars are read here,
    not at import time, so this works correctly regardless of when
    load_dotenv() runs relative to this module being imported). Caches
    the connection (or the failure reason) for the lifetime of the
    process.
    """

    global _client, _collection, _connection_error, _attempted

    if _attempted:
        return _collection

```

```

    _attempted = True

if not PYMONGO_AVAILABLE:
    _connection_error = "pymongo is not installed."
    return None

mongodb_uri = os.environ.get("MONGODB_URI")
mongodb_db = os.environ.get("MONGODB_DB", "mallinsight")

if not mongodb_uri:
    _connection_error = "MONGODB_URI is not configured."
    return None

try:
    _client = MongoClient(mongodb_uri, serverSelectionTimeoutMS=5000)
    _client.admin.command("ping")
    _collection = _client[mongodb_db]["analyses"]
except PyMongoError as e:
    _connection_error = str(e)
    return None

return _collection

def _reset_cache():
    """Test-only hook to force the next call to reconnect from scratch."""

    global _client, _collection, _connection_error, _attempted
    _client = None
    _collection = None
    _connection_error = None
    _attempted = False

def is_available():
    _get_collection()
    return _connection_error is None

def get_connection_error():
    _get_collection()
    return _connection_error

def save_analysis(metadata, hashes, score_info):

```

```
"""
```

Save a summary of one analysis run to MongoDB Atlas.
Returns the inserted record's id as a string, or None if the
database isn't configured/reachable (the caller should treat this
as non-fatal).

```
"""
```

```
collection = _get_collection()
```

```
if collection is None:
```

```
    return None
```

```
record = {
```

```
    "filename": metadata.get("name"),
```

```
    "size_kb": metadata.get("size"),
```

```
    "md5": hashes.get("md5"),
```

```
    "sha1": hashes.get("sha1"),
```

```
    "sha256": hashes.get("sha256"),
```

```
    "risk_score": score_info.get("score"),
```

```
    "verdict": score_info.get("verdict"),
```

```
    "reasons": score_info.get("reasons", []),
```

```
    "analyzed_at": datetime.now(timezone.utc),
```

```
}
```

```
try:
```

```
    result = collection.insert_one(record)
```

```
    return str(result.inserted_id)
```

```
except PyMongoError:
```

```
    return None
```

```
def get_recent_analyses(limit=50):
```

```
"""
```

Return the most recent analysis records, newest first.

Returns an empty list if the database isn't configured/reachable.

```
"""
```

```
collection = _get_collection()
```

```
if collection is None:
```

```
    return []
```

```
try:
```

```
    cursor = collection.find().sort("analyzed_at", -1).limit(limit)
```

```
    records = []
```

```
for doc in cursor:
    doc["id"] = str(doc.pop("_id"))
    records.append(doc)

return records
except PyMongoError:
    return []
```

YARA Rules

yara_rules/suspicious_indicators.yar

```
import "math"
```

```
rule Suspicious_Process_Injection
```

```
{
```

```
    meta:
```

```
        description = "References APIs commonly used for process injection"
```

```
        severity = "HIGH"
```

```
    strings:
```

```
        $a1 = "CreateRemoteThread" ascii wide
```

```
        $a2 = "WriteProcessMemory" ascii wide
```

```
        $a3 = "VirtualAllocEx" ascii wide
```

```
        $a4 = "NtUnmapViewOfSection" ascii wide
```

```
        $a5 = "QueueUserAPC" ascii wide
```

```
    condition:
```

```
        2 of ($a*)
```

```
}
```

```
rule Suspicious_Dynamic_API_Resolution
```

```
{
```

```
    meta:
```

```
        description = "Resolves APIs dynamically, often used to evade static detection"
```

```
        severity = "MEDIUM"
```

```
    strings:
```

```
        $a1 = "LoadLibraryA" ascii wide
```

```
        $a2 = "LoadLibraryW" ascii wide
```

```
        $a3 = "GetProcAddress" ascii wide
```

```
    condition:
```

```
        all of them
```

```
}
```

```
rule Suspicious_Command_Execution
```

```
{
```

```
    meta:
```

```
        description = "Contains references to command interpreters / script hosts"
```

```
        severity = "MEDIUM"
```

```
    strings:
```

```
$a1 = "cmd.exe" ascii wide nocase
$a2 = "powershell.exe" ascii wide nocase
$a3 = "wscript.exe" ascii wide nocase
$a4 = "mshta.exe" ascii wide nocase
$a5 = "WinExec" ascii
```

```
condition:
    any of them
}
```

```
rule Suspicious_AntiDebug_AntiVM
{
    meta:
        description = "Contains anti-debugging or anti-VM checks"
        severity = "MEDIUM"

    strings:
        $a1 = "IsDebuggerPresent" ascii
        $a2 = "CheckRemoteDebuggerPresent" ascii
        $a3 = "VMware" ascii nocase
        $a4 = "VBox" ascii nocase
        $a5 = "Sandboxie" ascii nocase
```

```
condition:
    any of them
}
```

```
rule Packed_File_High_Entropy
{
    meta:
        description = "Overall file entropy is very high, indicative of packing/encryption"
        severity = "MEDIUM"

    condition:
        filesize > 256 and math.entropy(0, filesize) >= 7.5
}
```

Templates (HTML)

templates/base.html

```
<!DOCTYPE html>
<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport"
        content="width=device-width, initial-scale=1.0">

    <title>{% block title %}MallnSight{% endblock %}</title>

    <!-- Google Font -->

    <link rel="preconnect"
        href="https://fonts.googleapis.com">

    <link rel="preconnect"
        href="https://fonts.gstatic.com"
        crossorigin>

    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&display=swap"
        rel="stylesheet">

    <!-- Bootstrap -->

    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
        rel="stylesheet">

    <!-- Bootstrap Icons -->

    <link rel="stylesheet"
        href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.3/font/bootstrap-icons.min.css">

    <!-- Tabler Icons -->

    <link rel="stylesheet"
        href="https://cdn.jsdelivr.net/npm/@tabler/icons-webfont@latest/tabler-icons.min.css">

    <!-- CSS -->
```



```
<link rel="stylesheet"
      href="{{ url_for('static', filename='css/style.css') }}">

</head>

<body>

<!-- NAVBAR -->

<nav class="navbar navbar-expand-lg navbar-dark fixed-top">

<div class="container">

<a class="navbar-brand"
  href="/">

<span class="logo-icon">

<i class="ti ti-shield-lock"></i>

</span>

MallnSight

</a>

<button class="navbar-toggler"
  type="button"
  data-bs-toggle="collapse"
  data-bs-target="#navbar">

<span class="navbar-toggler-icon"></span>

</button>

<div class="collapse navbar-collapse"
  id="navbar">

<ul class="navbar-nav ms-auto align-items-center">

<li class="nav-item">

<a class="nav-link {{ 'active' if request.path == '/' }}"
  href="/">
```

Home

<li class="nav-item">

<a class="nav-link {{ 'active' if request.path == '/features' }}"
href="/features">

Features

<li class="nav-item">

<a class="nav-link {{ 'active' if request.path == '/about' }}"
href="/about">

Documentation

<li class="nav-item">

<a class="nav-link {{ 'active' if request.path == '/history' }}"
href="/history">

History

<li class="nav-item">

<a class="nav-link {{ 'active' if request.path == '/contact' }}"
href="/contact">

Contact

<li class="nav-item">

<a class="nav-link"
href="https://github.com/vibushasatheeshkumar/mallnsight"
target="_blank" rel="noopener">

GitHub

<li class="nav-item ms-3">

<a class="btn btn-danger"

href="/upload">

Analyze File

</div>

</div>

</nav>

<!-- PAGE -->

<main>

{% block content %}

{% endblock %}

</main>

<!-- FOOTER -->

<footer>

<div class="container">

<div class="row">

<div class="col-lg-6">

<h5>

MallnSight

</h5>

<p>

Static Malware Analysis &
Threat Intelligence Platform.

</p>

<p class="footer-small">

A VivarX Project

</p>

</div>

<div class="col-lg-3">

<h6>

Platform

</h6>

Features

Analyze

</div>

<div class="col-lg-3">

<h6>

Resources

</h6>

Contact

GitHub

LinkedIn

</div>

</div>

<hr>

<div class="footer-bottom">

© 2026 MallnSight · A VivarX Project

</div>

</div>

</footer>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>

<script src="{{ url_for('static',filename='js/main.js') }}"></script>

</body>

</html>

templates/home.html

{% extends "base.html" %}

{% block title %}

MallnSight

{% endblock %}

{% block content %}

<!-- =====

HERO

===== -->

<section class="hero">

<div class="container">

<div class="row align-items-center">

<div class="col-lg-6">

<div class="project-badge">

A

VivarX Project

</div>

<h1>

MallnSight

</h1>

<h2>

Static Malware Analysis &

Threat Intelligence Platform

</h2>

<p>

Analyze suspicious files safely without executing them.
Perform enterprise-grade static malware analysis using
hashing, PE parsing, entropy calculation, YARA detection,
and automated investigation reports.

</p>

<div class="feature-grid">

<div class="feature-item">

<i class="ti ti-fingerprint"></i>

<div>

<h6>Hash Analysis</h6>

MD5 • SHA1 • SHA256

</div>

</div>

<div class="feature-item">

<i class="ti ti-file-code"></i>

<div>

<h6>PE Analysis</h6>

Headers • Imports

</div>

</div>

<div class="feature-item">

<i class="ti ti-shield-search"></i>

<div>

<h6>YARA Rules</h6>

Threat Detection

</div>

</div>

<div class="feature-item">

<i class="ti ti-chart-histogram"></i>

<div>

<h6>Entropy</h6>

Packed Detection

</div>

</div>

</div>

<div class="hero-buttons">

<a href="/upload"
class="btn btn-danger btn-lg">

Start Investigation

<a href="/features"
class="btn btn-outline-light btn-lg">

Learn More

</div>

</div>

<div class="col-lg-6">

<div class="hero-image">

<i class="ti ti-shield-lock-filled"></i>

</div>

</div>

</div>

</div>

</section>

<!-- =====
FEATURES
===== -->

<section class="features-section">

<div class="container">

<div class="section-title">

<div class="section-tag">

PLATFORM CAPABILITIES

</div>

<h2>

Enterprise Static Analysis

</h2>

<p>

Everything required to investigate suspicious files
without executing them.

</p>

</div>

<div class="row g-4">

<div class="col-lg-4">

<div class="feature-card">

<i class="ti ti-fingerprint"></i>

<h4>

Hash Analysis

</h4>

<p>

Generate MD5, SHA1 and SHA256 hashes
for file fingerprinting.

</p>

</div>

</div>

<div class="col-lg-4">

<div class="feature-card">

<i class="ti ti-file-code"></i>

<h4>

PE Analysis

</h4>

<p>

Inspect executable headers, imports,
sections and entry points.

</p>

</div>

</div>

<div class="col-lg-4">

<div class="feature-card">

<i class="ti ti-shield-search"></i>

<h4>

YARA Detection

</h4>

<p>

Detect malware families using
enterprise-grade YARA signatures.

</p>

</div>

</div>

<div class="col-lg-4">

<div class="feature-card">

<i class="ti ti-chart-histogram"></i>

<h4>

Entropy Analysis

</h4>

<p>

Detect packed and obfuscated
executables through entropy.

</p>

</div>

</div>

<div class="col-lg-4">

<div class="feature-card">

<i class="ti ti-search"></i>

<h4>

String Extraction

</h4>

<p>

Extract URLs, registry keys,
commands and suspicious strings.

</p>

</div>

</div>

<div class="col-lg-4">

<div class="feature-card">

<i class="ti ti-file-description"></i>

<h4>

Investigation Report

</h4>

<p>

Generate a professional PDF report
with indicators and risk score.

</p>

</div>

</div>

</div>

</div>

</section>

<!-- =====
WORKFLOW
===== -->

<section class="workflow">

<div class="container">

<div class="section-title">

<div class="section-tag">

WORKFLOW

</div>

<h2>

How MallnSight Works

</h2>

<p>

Simple, safe and completely offline.

</p>

</div>

<div class="row g-4">

<div class="col-lg-3">

<div class="workflow-step">

<div class="workflow-number">

1

</div>

<h5>

Upload File

</h5>

<p>

Choose a suspicious executable
or document.

</p>

</div>

</div>

<div class="col-lg-3">

<div class="workflow-step">

<div class="workflow-number">

2

</div>

<h5>

Static Analysis

</h5>

<p>

Hashes, metadata,
PE analysis and strings.

</p>

</div>

</div>

<div class="col-lg-3">

<div class="workflow-step">

<div class="workflow-number">

3

</div>

<h5>

YARA Scan

</h5>

<p>

Detect known malware families using rules.

</p>

</div>

</div>

<div class="col-lg-3">

<div class="workflow-step">

<div class="workflow-number">

4

</div>

<h5>

Investigation Report

</h5>

<p>

Download a detailed PDF analysis report.

</p>

</div>

</div>

</div>

</div>


```
</section>

<!-- =====
CTA
===== -->

<section>

<div class="container">

<div class="cta">

<h2>

Ready to Investigate?

</h2>

<p>

Start analyzing suspicious files safely
using MallnSight.

</p>

<a href="/upload"
class="btn btn-danger btn-lg">

Analyze File

</a>

</div>

</div>

</section>

{% endblock %}
```

templates/features.html

```
{% extends "base.html" %}

{% block title %}Features | MallnSight{% endblock %}
```

```
{% block content %}
```

```
<section class="page-hero">
```

```
<div class="container text-center">
```

```
<div class="section-tag">PLATFORM CAPABILITIES</div>
```

```
<h1 class="mt-3">Everything You Need for Static Analysis</h1>
```

```
<p class="text-secondary mt-3 mx-auto" style="max-width:720px;">
```

```
MallnSight inspects suspicious files completely offline &mdash; no code is  
ever executed. Every capability below runs locally in your own environment.
```

```
</p>
```

```
</div>
```

```
</section>
```

```
<section class="features-section pt-0">
```

```
<div class="container">
```

```
<div class="row g-4">
```

```
<div class="col-lg-4 col-md-6">
```

```
<div class="feature-card">
```

```
<i class="ti ti-fingerprint"></i>
```

```
<h4>Hash Analysis</h4>
```

```
<p>
```

```
Generates MD5, SHA1 and SHA256 hashes for unique file fingerprinting,  
useful for threat-intel lookups and de-duplication.
```

```
</p>
```

```
</div>
```

```
</div>
```

```
<div class="col-lg-4 col-md-6">
```

```
<div class="feature-card">
```

```
<i class="ti ti-file-info"></i>
```

```
<h4>File Metadata</h4>
```

```
<p>
```

```
Extracts filename, size, extension and MIME type to quickly  
characterize an unknown sample.
```

```
</p>
```

```
</div>
```

```
</div>
```

<div class="col-lg-4 col-md-6">

<div class="feature-card">

<i class="ti ti-file-code"></i>

<h4>PE Analysis</h4>

<p>

Parses Windows PE headers — architecture, compile timestamp, entry point, image base, sections, imports and exports — via `pefile`.

</p>

</div>

</div>

<div class="col-lg-4 col-md-6">

<div class="feature-card">

<i class="ti ti-chart-histogram"></i>

<h4>Entropy Analysis</h4>

<p>

Calculates whole-file and per-section Shannon entropy to flag packed, encrypted or obfuscated binaries.

</p>

</div>

</div>

<div class="col-lg-4 col-md-6">

<div class="feature-card">

<i class="ti ti-search"></i>

<h4>String Extraction</h4>

<p>

Pulls ASCII and UTF-16LE strings and automatically flags suspicious indicators — injection APIs, command interpreters, registry persistence, URLs, IPs and anti-debug checks.

</p>

</div>

</div>

<div class="col-lg-4 col-md-6">

<div class="feature-card">

<i class="ti ti-shield-search"></i>

<h4>YARA Detection</h4>

<p>

Scans against a bundled, extensible YARA rule set covering process injection, dynamic API resolution, command execution and packing.

</p>

</div>

</div>

```
<div class="col-lg-4 col-md-6">
<div class="feature-card">
<i class="ti ti-gauge"></i>
<h4>Risk Scoring</h4>
<p>
Combines every signal above into a single 0&ndash;100 risk score and
a clear verdict: Clean, Low Risk, Suspicious or High Risk.
</p>
</div>
</div>
```

```
<div class="col-lg-4 col-md-6">
<div class="feature-card">
<i class="ti ti-file-description"></i>
<h4>PDF Investigation Report</h4>
<p>
One-click export of the full findings into a shareable PDF report,
generated with <code>reportlab</code>.
</p>
</div>
</div>
```

```
<div class="col-lg-4 col-md-6">
<div class="feature-card">
<i class="ti ti-lock"></i>
<h4>Fully Offline</h4>
<p>
Nothing is uploaded to a third party. Files, hashes and results stay
on the machine running MallnSight.
</p>
</div>
</div>
```

```
</div>
```

```
</div>
</section>
```

```
<section>
<div class="container">
<div class="cta">
<h2>See It In Action</h2>
<p>Upload a file and get a full static analysis report in seconds.</p>
<a href="/upload" class="btn btn-danger btn-lg">Analyze File</a>
```

```
</div>
</div>
</section>
```

```
{% endblock %}
```

templates/about.html

```
{% extends "base.html" %}
```

```
{% block title %}Documentation | MallnSight{% endblock %}
```

```
{% block content %}
```

```
<section class="page-hero">
<div class="container text-center">
```

```
<div class="section-tag">DOCUMENTATION</div>
```

```
<h1 class="mt-3">MallnSight Documentation</h1>
```

```
<p class="text-secondary mt-3 mx-auto" style="max-width:720px;">
A complete reference for what MallnSight does, how it's built, and how
to run it yourself.
</p>
```

```
<a href="https://github.com/vibushasatheeshkumar/mallnsight"
  target="_blank" rel="noopener"
  class="btn btn-outline-light mt-4">
  <i class="ti ti-brand-github"></i>
  View Source on GitHub
</a>
```

```
</div>
</section>
```

```
<section class="docs-section">
<div class="container">
```

```
<div class="row g-5">
```

```
<!-- TOC -->
<div class="col-lg-3">
<nav class="doc-toc">
<h6>On this page</h6>
```

```
<a href="#overview">Overview</a>
<a href="#tech-stack">Tech Stack</a>
<a href="#structure">Project Structure</a>
<a href="#getting-started">Getting Started</a>
<a href="#usage">Usage</a>
<a href="#yara">Extending YARA Rules</a>
<a href="#deployment">Deployment</a>
<a href="#disclaimer">Disclaimer</a>
</nav>
</div>
```

```
<!-- CONTENT -->
```

```
<div class="col-lg-9">
```

```
<div class="doc-block" id="overview">
```

```
<h3>Overview</h3>
```

```
<p>
```

MallnSight is a Flask-based web application for performing
offline, static analysis of suspicious files. It never
executes the uploaded file — it inspects hashes, PE structure,
entropy, embedded strings and YARA signatures to produce a risk score
and a downloadable PDF investigation report.

```
</p>
```

<p>All analysis runs locally; no file or hash is ever sent to a third-party service.</p>

```
</div>
```

```
<div class="doc-block" id="tech-stack">
```

```
<h3>Tech Stack</h3>
```

```
<ul class="doc-list">
```

```
<li><strong>Backend:</strong> Python, Flask</li>
```

```
<li><strong>PE Parsing:</strong> pefile</li>
```

```
<li><strong>Pattern Matching:</strong> yara-python</li>
```

```
<li><strong>PDF Generation:</strong> reportlab</li>
```

```
<li><strong>Frontend:</strong> Jinja2, Bootstrap 5, vanilla JS</li>
```

```
<li><strong>CI:</strong> GitHub Actions (pytest on Python 3.11 / 3.12)</li>
```

```
</ul>
```

```
</div>
```

```
<div class="doc-block" id="structure">
```

```
<h3>Project Structure</h3>
```

```
<pre class="doc-code"><code>mallnsight/
```

```
■■■■ app.py # Flask routes & request handling
```

```
■■■■ analysis/
```

```
■ ■■■■ hash.py # MD5 / SHA1 / SHA256
```

```
■ ■■■■ metadata.py # File info (size, MIME, extension)
```

- ■■■■ pe_analysis.py # PE header / sections / imports / exports
- ■■■■ entropy.py # Shannon entropy
- ■■■■ strings.py # String extraction + suspicious indicators
- ■■■■ yara_scan.py # YARA rule compilation & scanning
- ■■■■ scoring.py # Risk score & verdict
- ■■■■ report.py # PDF report generation
- yara_rules/ # Bundled .yar rule files
- templates/ # Jinja2 pages
- static/ # CSS / JS / assets
- tests/ # pytest smoke tests
- uploads/ # Uploaded files (runtime only)
- reports/ # Generated PDF reports (runtime only)</code></pre>

Getting Started

Prerequisites

- Python 3.11 or 3.12 (prebuilt `yara-python` wheels are available for these versions)
- On Windows, building `yara-python` from source requires the [Microsoft C++ Build Tools](https://visualstudio.microsoft.com/visual-cpp-build-tools/)

Setup

```
class="doc-code">git clone git@github.com:vibushasatheeshkumar/mallInsight.git
cd mallInsight
```

```
python -m venv venv
venv\Scripts\activate # Windows
source venv/bin/activate # macOS/Linux
```

```
pip install -r requirements.txt</code></pre>
```

Run

```
class="doc-code">python app.py</code></pre>

The app starts at http://127.0.0.1:5000. Debug mode is off by default; set FLASK_DEBUG=1 to enable it during development.


```

Usage

- Go to `/upload`.

- Drag and drop, or browse, a file to analyze (`.exe`, `.dll`, `.sys`, `.msi`, `.zip`, `.apk`, `.pdf`, `.docx`, `.bin`).
- View the generated investigation dashboard: risk score, hashes, PE breakdown, entropy chart, suspicious strings, and YARA matches.
- Download the findings as a PDF report.

Uploaded files are size-limited to 100MB and stored under a randomly generated filename to avoid collisions.

</div>

Extending YARA Rules</h3>

<p>

Drop additional `.yar` files into `yara_rules/` — they're compiled and scanned automatically on the next analysis request.

Each rule should set a `severity` meta field

(`HIGH`, `MEDIUM`, or `LOW`), which feeds directly into the risk score.

</p>

</div>

</div>

Deployment</h3>

<p>

MallnSight ships ready to deploy on [Render](https://render.com)'s free tier:

</p>

- The repo includes a `Procfile` (`waitress-serve --host=0.0.0.0 --port=$PORT app:app`) and a `runtime.txt` pinning Python 3.12.

- On Render: New → Web Service → connect this GitHub repo → Build Command `pip install -r requirements.txt` → Start Command is picked up automatically from the `Procfile`.

- The free tier's filesystem is ephemeral — uploaded files and generated reports won't persist across restarts, which is expected for a stateless demo deployment.

</div>

</div>

Disclaimer</h3>

<p>

MallnSight performs **static** analysis only — it does not execute or detonate uploaded files. It is intended for educational and authorized security research/testing purposes. Always handle suspicious

files in accordance with your organization's security policy.

</p>

</div>

</div>

</div>

</div>

</section>

{% endblock %}

templates/upload.html

{% extends "base.html" %}

{% block title %}Analyze File | MallnSight{% endblock %}

{% block content %}

<section class="upload-page">

<div class="container">

<!-- Header -->

<div class="row justify-content-center">

<div class="col-lg-8 text-center">

OFFLINE STATIC ANALYSIS

<h1 class="display-4 fw-bold mt-3">

Analyze Suspicious Files

</h1>

<p class="text-secondary mt-4">

Upload suspicious files for enterprise-grade malware analysis.

MallnSight performs completely offline static inspection using

hashing, PE analysis, string extraction, entropy analysis and YARA detection.

</p>

</div>

</div>

<!-- Upload Card -->

<div class="row justify-content-center mt-5">

<div class="col-lg-9">

<form

action="/analyze"

method="POST"

enctype="multipart/form-data">

<div class="upload-card">

<div class="upload-icon">

<i class="ti ti-cloud-upload"></i>

</div>

<h2>

Drag & Drop Your File

</h2>

<p class="text-secondary mt-3">

or browse a file from your computer

</p>

<input

type="file"

id="fileInput"

name="file"

hidden>

```
<label
  for="fileInput"
  class="browse-btn mt-4">

  <i class="ti ti-folder-open"></i>
```

Browse File

```
</label>
```

```
<div
  id="filename"
  class="selected-file mt-4">
```

No file selected

```
</div>
```

```
<div class="supported-files mt-5">
```

```
<h6>
```

Supported Formats

```
</h6>
```

```
<div class="mt-3">
```

```
<span class="file-pill">EXE</span>
```

```
<span class="file-pill">DLL</span>
```

```
<span class="file-pill">MSI</span>
```

```
<span class="file-pill">ZIP</span>
```

```
<span class="file-pill">APK</span>
```

```
<span class="file-pill">PDF</span>
```

```
<span class="file-pill">DOCX</span>
```

```
</div>
```

```

        </div>

        <button
            type="submit"
            class="analyze-btn mt-5">

            <i class="ti ti-shield-search"></i>

            Start Investigation

        </button>

    </div>

</form>

</div>

</div>

</div>

</section>

<script>

const fileInput=document.getElementById("fileInput");

const filename=document.getElementById("filename");

fileInput.addEventListener("change",()=>{

    if(fileInput.files.length>0){

        filename.innerHTML=fileInput.files[0].name;

    }

});

</script>

{% endblock %}

/* =====
    UPLOAD PAGE

```

```
===== */
```

```
.upload-page{
```

```
    padding-top:140px;
```

```
    padding-bottom:120px;
```

```
}
```

```
/* Upload Card */
```

```
.upload-card{
```

```
    background:#161B22;
```

```
    border:1px solid #30363D;
```

```
    border-radius:20px;
```

```
    padding:70px 50px;
```

```
    text-align:center;
```

```
    transition:.35s;
```

```
}
```

```
.upload-card:hover{
```

```
    border-color:#E11D48;
```

```
    transform:translateY(-5px);
```

```
    box-shadow:0 20px 45px rgba(225,29,72,.12);
```

```
}
```

```
/* Upload Icon */
```

```
.upload-icon{
```

```
    width:110px;
```

```
    height:110px;
```

```
    margin:auto;
```

```
    border-radius:50%;
```

```
display:flex;
align-items:center;
justify-content:center;

background:#1C2128;

color:#E11D48;

font-size:50px;

margin-bottom:30px;

animation:uploadPulse 3s ease-in-out infinite;
}

@keyframes uploadPulse{

  0%,100%{

    transform:scale(1);

    box-shadow:0 0 0 rgba(225,29,72,0);

  }

  50%{

    transform:scale(1.08);

    box-shadow:0 0 25px rgba(225,29,72,.35);

  }

}

/* Browse Button */

.browse-btn{

display:inline-flex;

align-items:center;
```

```
    gap:10px;

    padding:14px 30px;

    border-radius:10px;

    background:#E11D48;

    color:white;

    font-weight:600;

    cursor:pointer;

    transition:.3s;

    text-decoration:none;
}

.browse-btn:hover{

    background:#F43F5E;

    transform:translateY(-3px);

    color:white;
}

/* Selected File */

.selected-file{

    color:#CBD5E1;

    font-weight:600;
}

/* Supported Files */

.supported-files h6{

    color:#9AA3B2;
```

```
    font-size:14px;

    letter-spacing:1px;
}

.file-pill{

    display:inline-block;

    margin:6px;

    padding:8px 16px;

    background:#1C2128;

    border:1px solid #30363D;

    border-radius:30px;

    color:#CBD5E1;

    transition:.3s;
}

.file-pill:hover{

    border-color:#E11D48;

    transform:translateY(-2px);
}

/* Analyze Button */

.analyze-btn{

    border:none;

    background:#E11D48;

    color:white;
```



```
border-radius:10px;

padding:15px 38px;

display:inline-flex;

align-items:center;

justify-content:center;

gap:12px;

font-size:16px;

font-weight:600;

transition:.35s;

position:relative;

overflow:hidden;
}

.analyze-btn::before{

content:"";

position:absolute;

top:0;

left:-120%;

width:40%;

height:100%;

background:linear-gradient(

    90deg,

    transparent,

    rgba(255,255,255,.45),
```

```
        transparent

    );

    transform:skewX(-20deg);

    animation:scanButton 3s linear infinite;

}

@keyframes scanButton{

    100%{

        left:150%;

    }

}

.analyze-btn:hover{

    background:#F43F5E;

    transform:translateY(-3px);

    box-shadow:0 15px 35px rgba(225,29,72,.30);

}

/* Responsive */

@media(max-width:768px){

    .upload-card{

        padding:40px 25px;

    }

    .upload-icon{

        width:80px;
```

```
    height:80px;

    font-size:36px;

}

.analyze-btn{

    width:100%;

}

.browse-btn{

    width:100%;

    justify-content:center;

}

}

/* =====
SECURITY SECTION
===== */

.security-section{

    margin-top:90px;

}

.security-card{

    background:#161B22;

    border:1px solid #30363D;

    border-radius:18px;

    padding:35px;

    text-align:center;

    height:100%;
```

```
    transition:.35s ease;

}

.security-card:hover{

    transform:translateY(-8px);

    border-color:#E11D48;

    box-shadow:0 18px 40px rgba(225,29,72,.12);

}

.security-icon{

    width:70px;

    height:70px;

    margin:auto;

    margin-bottom:22px;

    border-radius:50%;

    background:#1C2128;

    display:flex;

    align-items:center;

    justify-content:center;

    transition:.35s;

}

.security-card:hover .security-icon{

    background:#E11D48;

}

.security-icon i{
```

```
font-size:30px;

color:#E11D48;

transition:.35s;

}

.security-card:hover .security-icon i{

color:white;

}

.security-card h4{

font-size:22px;

font-weight:700;

margin-bottom:15px;

}

.security-card p{

color:#9AA3B2;

line-height:1.8;

}

/* =====
WHY MALLNSIGHT
===== */

.investigation-info{

margin-top:70px;

background:#161B22;

border:1px solid #30363D;
```

```
border-radius:22px;

padding:60px;

text-align:center;
}
```

```
.investigation-info h3{

font-size:34px;

font-weight:700;

margin-bottom:18px;
}
```

```
.investigation-info p{

max-width:760px;

margin:auto;

color:#9AA3B2;

line-height:1.9;
}
```

```
.info-tags{

margin-top:35px;
}
```

```
.info-tags span{

display:inline-block;

margin:6px;

padding:10px 18px;

border-radius:30px;
```

```

border:1px solid #30363D;

background:#1C2128;

color:#CBD5E1;

transition:.3s;
}

.info-tags span:hover{

border-color:#E11D48;

color:white;

transform:translateY(-2px);
}

/* =====
DRAG & DROP EFFECT
===== */

.upload-card.dragover{

border:2px dashed #E11D48;

background:#1A1F28;

transform:scale(1.02);

box-shadow:0 0 35px rgba(225,29,72,.25);
}

/* =====
FILE NAME
===== */

.selected-file{

display:inline-block;

```

```

margin-top:25px;

padding:10px 18px;

border-radius:30px;

background:#1C2128;

border:1px solid #30363D;

color:#CBD5E1;

}

/* =====
SMOOTH ANIMATION
===== */

.upload-card,
.security-card,
.investigation-info{

    animation:fadeUp .8s ease;

}

@keyframes fadeUp{

    from{

        opacity:0;

        transform:translateY(30px);

    }

    to{

        opacity:1;

        transform:translateY(0);

    }

}

```


templates/dashboard.html

```
{% extends "base.html" %}

{% block title %}Investigation Report | MallnSight{% endblock %}

{% block content %}

<section class="dashboard-page">
<div class="container">

{% if not metadata %}

<div class="error-box">
    No analysis results yet. <a href="/upload">Upload a file</a> to get started.
</div>

{% else %}

{% set risk_class = {
    "HIGH RISK": "risk-high",
    "SUSPICIOUS": "risk-medium",
    "LOW RISK": "risk-low",
    "CLEAN": "risk-clean"
}.get(score_info.verdict, "risk-low") %}

<!-- =====
VERDICT
===== -->

<div class="verdict-card">
<div class="row align-items-center">

<div class="col-lg-8">

<span class="verdict-tag {{ risk_class }}">
    {{ score_info.verdict }}
</span>

<h2 class="mt-3">
    {{ metadata.name }}
</h2>

<p>
    This file was analyzed offline using hashing, PE inspection, entropy
    analysis, string extraction and YARA pattern matching.
```

</p>

```
{% if score_info.reasons %}
<ul class="text-secondary">
  {% for reason in score_info.reasons %}
    <li>{{ reason }}</li>
  {% endfor %}
</ul>
{% endif %}
```

</div>

<div class="col-lg-4 text-end">

```
<div class="risk-circle {{ risk_class }}">
{{ score_info.score }}%
</div>
```

```
<span class="risk-label">
Risk Score
</span>
```

</div>

</div>

</div>

```
<!-- =====
FILE INFO
===== -->
```

<div class="row mt-5 g-4">

```
<div class="col-lg-6">
<div class="report-card">
<h4>File Information</h4>
<table class="table table-dark">
<tr><td>Filename</td><td>{{ metadata.name }}</td></tr>
<tr><td>File Size</td><td>{{ metadata.size }} KB</td></tr>
<tr><td>Extension</td><td>{{ metadata.extension }}</td></tr>
<tr><td>MIME</td><td>{{ metadata.mime or "Unknown" }}</td></tr>
</table>
</div>
</div>
```

```
<div class="col-lg-6">
<div class="report-card">
<h4>Hashes</h4>
<table class="table table-dark">
<tr>
<td>MD5</td>
<td>
<span class="hash-value">{{ hashes.md5 }}</span>
<button class="copy-btn" data-copy="{{ hashes.md5 }}">Copy</button>
</td>
</tr>
<tr>
<td>SHA1</td>
<td>
<span class="hash-value">{{ hashes.sha1 }}</span>
<button class="copy-btn" data-copy="{{ hashes.sha1 }}">Copy</button>
</td>
</tr>
<tr>
<td>SHA256</td>
<td>
<span class="hash-value">{{ hashes.sha256 }}</span>
<button class="copy-btn" data-copy="{{ hashes.sha256 }}">Copy</button>
</td>
</tr>
</table>
</div>
</div>

</div>
```

```
<!-- =====
PE ANALYSIS
===== -->
```

```
<div class="row mt-4">
<div class="col-lg-12">
<div class="report-card">
<h4>PE Analysis</h4>
```

```
{% if pe_info.error %}
```

```
<div class="error-box">
    Not a recognized PE file ({{ pe_info.error }}). PE-specific analysis
    (imports, exports, sections) is unavailable for this file type.
```

</div>

{% else %}

<table class="table table-dark">

<thead>

<tr><th>Property</th><th>Value</th></tr>

</thead>

<tbody>

<tr><td>Architecture</td><td>{{ pe_info.architecture }}</td></tr>

<tr><td>Compile Time</td><td>{{ pe_info.compile_time }}</td></tr>

<tr><td>Entry Point</td><td>{{ pe_info.entry_point }}</td></tr>

<tr><td>Image Base</td><td>{{ pe_info.image_base }}</td></tr>

<tr><td>Number of Sections</td><td>{{ pe_info.number_of_sections }}</td></tr>

</tbody>

</table>

<h5 class="mt-4">Sections</h5>

<table class="table table-dark">

<thead>

<tr><th>Name</th><th>Virtual Size</th><th>Raw Size</th><th>Entropy</th></tr>

</thead>

<tbody>

{% for section in pe_info.sections %}

<tr>

<td>{{ section.name }}</td>

<td>{{ section.virtual_size }}</td>

<td>{{ section.raw_size }}</td>

<td>{{ section.entropy }}</td>

</tr>

{% endfor %}

</tbody>

</table>

<h5 class="mt-4">Imports ({{ pe_info.imports|length }})</h5>

{% if pe_info.imports %}

<table class="table table-dark">

<thead>

<tr><th>DLL</th><th>Function</th></tr>

</thead>

<tbody>

{% for imp in pe_info.imports[:50] %}

<tr><td>{{ imp.dll }}</td><td>{{ imp.function }}</td></tr>

{% endfor %}

</tbody>

```
</table>
{% if pe_info.imports|length > 50 %}
<small class="text-secondary">Showing first 50 of {{ pe_info.imports|length }} imports.</small>
{% endif %}
{% else %}
<p class="text-secondary">No imports found.</p>
{% endif %}
```

```
<h5 class="mt-4">Exports ({{ pe_info.exports|length }})</h5>
{% if pe_info.exports %}
<ul>
{% for exp in pe_info.exports %}
<li>{{ exp }}</li>
{% endfor %}
</ul>
{% else %}
<p class="text-secondary">No exported functions found.</p>
{% endif %}
```

```
{% endif %}
```

```
</div>
```

```
</div>
```

```
</div>
```

```
<!-- =====
YARA
===== -->
```

```
<div class="row mt-4">
```

```
<div class="col-lg-6">
```

```
<div class="report-card">
```

```
<h4>YARA Matches</h4>
```

```
{% if not yara_info.available %}
```

```
<div class="error-box">
```

```
YARA scanning unavailable: {{ yara_info.error }}
```

```
</div>
```

```
{% elif yara_info.matches %}
```

```
<table class="table table-dark">
```

```
<thead>
```

```

<tr><th>Rule</th><th>Severity</th></tr>
</thead>
<tbody>
{% for match in yara_info.matches %}
<tr>
<td>{{ match.rule }}</td>
<td>
{% if match.severity == "HIGH" %}
<span class="badge bg-danger">HIGH</span>
{% elif match.severity == "MEDIUM" %}
<span class="badge bg-warning text-dark">MEDIUM</span>
{% else %}
<span class="badge bg-secondary">LOW</span>
{% endif %}
</td>
</tr>
{% endfor %}
</tbody>
</table>

```

```
{% else %}
```

```
<p class="text-secondary">No YARA rules matched.</p>
```

```
{% endif %}
```

```
</div>
```

```
</div>
```

```

<!-- =====
ENTROPY
===== -->

```

```

<div class="col-lg-6">
<div class="report-card">
<h4>File Entropy</h4>

```

```

<div class="entropy-item">
<span>Overall ({{ entropy_info.verdict }})</span>
<div class="progress">
{% set pct = (entropy_info.entropy / 8 * 100)|round(1) %}
<div class="progress-bar {% if entropy_info.entropy >= 7.5 %}bg-danger{% elif entropy_info.entropy >= 6.5
%}bg-warning{% else %}bg-success{% endif %}" style="width:{{ pct }}%">
</div>
</div>

```

```
<small>{{ entropy_info.entropy }} / 8.0</small>
</div>
```

```
{% if not pe_info.error %}
{% for section in pe_info.sections %}
<div class="entropy-item mt-4">
<span>{{ section.name }}</span>
<div class="progress">
{% set spct = (section.entropy / 8 * 100)|round(1) %}
<div class="progress-bar {% if section.entropy >= 7.5 %}bg-danger{% elif section.entropy >= 6.5
%}bg-warning{% else %}bg-success{% endif %}" style="width:{{ spct }}%">
</div>
</div>
<small>{{ section.entropy }} / 8.0</small>
</div>
{% endfor %}
{% endif %}
```

```
</div>
</div>
```

```
</div>
```

```
<!-- =====
STRINGS
===== -->
```

```
<div class="row mt-4">
<div class="col-lg-12">
<div class="report-card">
<h4>Suspicious Strings ({{ strings_info.suspicious|length }})</h4>
```

```
{% if strings_info.suspicious %}
```

```
<div class="strings-box">
{% for s in strings_info.suspicious %}
<span class="string-pill" title="{{ s.category }}">{{ s.value }}</span>
{% endfor %}
</div>
```

```
{% else %}
```

```
<p class="text-secondary">No suspicious strings detected.</p>
```

```
{% endif %}
```

```
<p class="text-secondary mt-3">
{{ strings_info.total_strings }} printable strings extracted in total.
</p>
```

```
</div>
</div>
</div>
```

```
<!-- =====
DOWNLOAD
===== -->
```

```
<div class="text-center mt-5">
<a href="/download/{{ download_id }}" class="btn btn-danger btn-lg download-btn">
<i class="ti ti-download"></i>
Download PDF Report
</a>
</div>
```

```
{% endif %}
```

```
</div>
</section>
```

```
<script>
document.querySelectorAll(".copy-btn").forEach(btn => {
  btn.addEventListener("click", () => {
    navigator.clipboard.writeText(btn.dataset.copy);
    const original = btn.textContent;
    btn.textContent = "Copied!";
    setTimeout(() => { btn.textContent = original; }, 1500);
  });
});
</script>
```

```
{% endblock %}
```

templates/history.html

```
{% extends "base.html" %}
```

```
{% block title %}History | MallnSight{% endblock %}
```

```
{% block content %}
```



```
<section class="page-hero">
```

```
<div class="container text-center">
```

```
<div class="section-tag">CLOUD ANALYSIS HISTORY</div>
```

```
<h1 class="mt-3">Recent Analyses</h1>
```

```
<p class="text-secondary mt-3 mx-auto" style="max-width:680px;">
```

```
Past results stored in MongoDB Atlas, so previously analyzed files stay  
searchable even after the local upload/report files are gone.
```

```
</p>
```

```
</div>
```

```
</section>
```

```
<section class="pt-0">
```

```
<div class="container">
```

```
{% if not available %}
```

```
<div class="error-box">
```

```
History is unavailable: {{ error }}.
```

```
See <code>DEVELOPER_GUIDE.md</code> for how to configure
```

```
<code>MONGODB_URI</code> to enable this feature.
```

```
</div>
```

```
{% elif not records %}
```

```
<div class="report-card text-center">
```

```
<p class="text-secondary mb-0">
```

```
No analyses recorded yet. Results are saved here automatically every  
time a file is analyzed.
```

```
</p>
```

```
</div>
```

```
{% else %}
```

```
<div class="report-card">
```

```
<table class="table table-dark">
```

```
<thead>
```

```
<tr>
```

```
<th>Analyzed At (UTC)</th>
```

```
<th>Filename</th>
```

```
<th>SHA256</th>
```

```

<th>Verdict</th>
<th>Risk Score</th>
</tr>
</thead>
<tbody>
{% for record in records %}
<tr>
<td>{{ record.analyzed_at.strftime("%Y-%m-%d %H:%M") if record.analyzed_at else "" }}</td>
<td>{{ record.filename }}</td>
<td><span class="hash-value">{{ record.sha256[:16] }}&hellip;</span></td>
<td>
{% if record.verdict == "HIGH RISK" %}
<span class="badge bg-danger">{{ record.verdict }}</span>
{% elif record.verdict == "SUSPICIOUS" %}
<span class="badge bg-warning text-dark">{{ record.verdict }}</span>
{% elif record.verdict == "LOW RISK" %}
<span class="badge bg-primary">{{ record.verdict }}</span>
{% else %}
<span class="badge bg-success">{{ record.verdict }}</span>
{% endif %}
</td>
<td>{{ record.risk_score }}/100</td>
</tr>
{% endfor %}
</tbody>
</table>
</div>

{% endif %}

</div>
</section>

{% endblock %}

```

templates/contact.html

```

{% extends "base.html" %}

{% block title %}Contact | MallnSight{% endblock %}

{% block content %}

<section class="page-hero">
<div class="container text-center">

```

```
<div class="section-tag">GET IN TOUCH</div>
```

```
<h1 class="mt-3">Contact</h1>
```

```
<p class="text-secondary mt-3 mx-auto" style="max-width:620px;">
Questions, bug reports, or ideas for MallnSight? Reach out directly
or open an issue on GitHub.
</p>
```

```
</div>
```

```
</section>
```

```
<section class="pt-0">
```

```
<div class="container">
```

```
<div class="row g-4 justify-content-center">
```

```
<div class="col-lg-4 col-md-6">
```

```
<a href="mailto:vibushasatheeshkumar@gmail.com" class="contact-card">
```

```
<i class="ti ti-mail"></i>
```

```
<h4>Email</h4>
```

```
<p>vibushasatheeshkumar@gmail.com</p>
```

```
</a>
```

```
</div>
```

```
<div class="col-lg-4 col-md-6">
```

```
<a href="https://github.com/vibushasatheeshkumar/mallnsight" target="_blank" rel="noopener"
class="contact-card">
```

```
<i class="ti ti-brand-github"></i>
```

```
<h4>GitHub</h4>
```

```
<p>vibushasatheeshkumar/mallnsight</p>
```

```
</a>
```

```
</div>
```

```
<div class="col-lg-4 col-md-6">
```

```
<a href="https://www.linkedin.com/in/vibushasatheeshkumar" target="_blank" rel="noopener"
class="contact-card">
```

```
<i class="ti ti-brand-linkedin"></i>
```

```
<h4>LinkedIn</h4>
```

```
<p>linkedin.com/in/vibushasatheeshkumar</p>
```

```
</a>
```

```
</div>
```

```
</div>
```

</div>

</section>

{% endblock %}

Static Assets

static/css/style.css

```
/* =====  
MALLNSIGHT ENTERPRISE UI  
===== */  
  
@import url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&display=swap');  
  
:root{  
  
    --bg:#0D1117;  
    --surface:#161B22;  
    --card:#1C2128;  
    --border:#30363D;  
  
    --primary:#E11D48;  
    --primary-hover:#F43F5E;  
  
    --text:#F8FAFC;  
    --muted:#9AA3B2;  
  
    --success:#22C55E;  
    --warning:#FACC15;  
    --danger:#EF4444;  
  
}  
  
*{  
    margin:0;  
    padding:0;  
    box-sizing:border-box;  
}  
  
html{  
  
    scroll-behavior:smooth;  
  
}  
  
body{  
  
    background:var(--bg);
```

```
color:var(--text);

font-family:"Inter",sans-serif;

overflow-x:hidden;

line-height:1.7;

}

/* =====
BACKGROUND
===== */

body::before{

content:"";

position:fixed;

inset:0;

background-image:

linear-gradient(rgba(255,255,255,.02) 1px,transparent 1px),

linear-gradient(90deg,rgba(255,255,255,.02) 1px,transparent 1px);

background-size:60px 60px;

opacity:.25;

pointer-events:none;

z-index:-2;

}

body::after{

content:"";

position:fixed;

inset:0;
```

```
background:

radial-gradient(circle at 20% 20%,rgba(255,255,255,.02),transparent 35%),

radial-gradient(circle at 80% 70%,rgba(255,255,255,.015),transparent 35%);

z-index:-1;

pointer-events:none;

}

/* =====
  NAVBAR
===== */

.navbar{

  background:rgba(13,17,23,.92);

  backdrop-filter:blur(12px);

  border-bottom:1px solid var(--border);

  padding:14px 0;

}

.navbar-brand{

  color:white!important;

  font-weight:700;

  font-size:24px;

  letter-spacing:.3px;

}

.logo-icon{

  color:var(--primary);
```

```
        margin-right:10px;

    }

    .nav-link{

        color:var(--muted)!important;

        margin-left:18px;

        font-weight:500;

        transition:.3s;

    }

    .nav-link:hover{

        color:white!important;

    }

    .nav-link.active{

        color:white!important;

    }

    /* =====
    BUTTONS
    ===== */

    .btn{

        border-radius:8px!important;

        transition:.3s;

    }

    .btn-danger{

        background:var(--primary);

        border:none;
```



```
}
```

```
.btn-danger:hover{
```

```
    background:var(--primary-hover);
```

```
    transform:translateY(-2px);
```

```
    box-shadow:0 12px 25px rgba(225,29,72,.20);
```

```
}
```

```
.btn-outline-light:hover{
```

```
    transform:translateY(-2px);
```

```
}
```

```
/* =====
```

```
HERO
```

```
===== */
```

```
.hero{
```

```
    min-height:92vh;
```

```
    display:flex;
```

```
    align-items:center;
```

```
    padding-top:90px;
```

```
}
```

```
.hero h1{
```

```
    font-size:74px;
```

```
    font-weight:800;
```

```
    margin-bottom:15px;
```

```
}
```

```
.hero h2{

  color:#9AA3B2;

  font-weight:400;

  font-size:28px;

  line-height:1.5;

}

.hero p{

  color:#94A3B8;

  font-size:18px;

  margin-top:30px;

  max-width:620px;

}

.hero-buttons{

  display:flex;

  gap:15px;

  margin-top:40px;

}

.project-badge{

  display:inline-flex;

  align-items:center;

  gap:8px;

  background:#161B22;

  border:1px solid var(--border);
```

```
    color:#9AA3B2;

    border-radius:25px;

    padding:8px 18px;

    font-size:13px;

    margin-bottom:20px;

}
```

```
.project-name{

    color:var(--primary);

    font-weight:700;

}
```

```
/* =====
    RIGHT ILLUSTRATION
    ===== */
```

```
.hero-image{

    display:flex;

    justify-content:flex-end;

    align-items:center;

}
```

```
.hero-image i{

    font-size:130px;

    color:var(--primary);

}
```

```
/* =====
    FEATURE GRID
```

```
===== */
```

```
.feature-grid{
```

```
  display:grid;
```

```
  grid-template-columns:repeat(2,1fr);
```

```
  gap:18px;
```

```
  margin-top:35px;
```

```
}
```

```
.feature-item{
```

```
  background:#161B22;
```

```
  border:1px solid var(--border);
```

```
  border-radius:14px;
```

```
  padding:18px;
```

```
  display:flex;
```

```
  align-items:center;
```

```
  gap:16px;
```

```
  transition:.35s;
```

```
}
```

```
.feature-item:hover{
```

```
  border-color:var(--primary);
```

```
  transform:translateY(-5px);
```

```
}
```

```
.feature-item i{
```

```
  font-size:24px;
```

```
    color:var(--primary);  
  
}
```

```
.feature-item h6{  
  
    margin:0;  
  
    font-weight:600;  
  
}
```

```
.feature-item span{  
  
    font-size:13px;  
  
    color:#94A3B8;  
  
}
```

```
/* =====  
SECTION  
===== */
```

```
section{  
  
    padding:110px 0;  
  
}
```

```
.section-title{  
  
    text-align:center;  
  
    margin-bottom:70px;  
  
}
```

```
.section-tag{  
  
    color:var(--primary);  
  
    text-transform:uppercase;
```

```
    letter-spacing:2px;

    font-size:13px;

    font-weight:600;
}

.section-title h2{

    font-size:44px;

    font-weight:700;

    margin-top:15px;
}

.section-title p{

    color:#9AA3B2;

    max-width:650px;

    margin:auto;

    margin-top:18px;
}

/* =====
FEATURE CARDS
===== */

.features-section{

    padding-top:40px;
}

.feature-card{

    background:var(--surface);

    border:1px solid var(--border);
```

```
border-radius:18px;

padding:35px;

height:100%;

transition:.35s ease;
}

.feature-card:hover{

transform:translateY(-8px);

border-color:var(--primary);

box-shadow:0 18px 40px rgba(225,29,72,.10);
}

.feature-card i{

font-size:38px;

color:var(--primary);

margin-bottom:25px;
}

.feature-card h4{

font-size:22px;

font-weight:700;

margin-bottom:15px;
}

.feature-card p{

color:var(--muted);
```

```
    font-size:15px;

    line-height:1.8;

}

/* =====
   WORKFLOW
   ===== */

.workflow{

    position:relative;

}

.workflow-step{

    background:var(--surface);

    border:1px solid var(--border);

    border-radius:16px;

    padding:30px;

    text-align:center;

    position:relative;

    transition:.3s;

}

.workflow-step:hover{

    border-color:var(--primary);

    transform:translateY(-6px);

}

.workflow-number{

    width:55px;
```



```
height:55px;

background:var(--primary);

border-radius:50%;

display:flex;

align-items:center;

justify-content:center;

margin:auto;

margin-bottom:20px;

font-weight:700;

font-size:20px;

}
```

```
.workflow-step h5{

margin-bottom:10px;

font-weight:700;

}
```

```
.workflow-step p{

color:var(--muted);

font-size:14px;

}
```

```
/* =====
CTA
===== */
```

```
.cta{
```

```
background:#161B22;

border:1px solid var(--border);

border-radius:20px;

text-align:center;

padding:70px 40px;

}

.cta h2{

    font-size:46px;

    font-weight:700;

}

.cta p{

    color:var(--muted);

    font-size:18px;

    margin:20px auto 35px;

    max-width:650px;

}

/* =====
FOOTER
===== */

footer{

    border-top:1px solid var(--border);

    margin-top:120px;

    padding:70px 0 30px;

    background:#0B0F14;
```

```
}
```

```
footer h5{
```

```
    font-weight:700;
```

```
    margin-bottom:15px;
```

```
}
```

```
footer h6{
```

```
    color:white;
```

```
    margin-bottom:15px;
```

```
}
```

```
footer p{
```

```
    color:var(--muted);
```

```
}
```

```
footer a{
```

```
    color:var(--muted);
```

```
    text-decoration:none;
```

```
    line-height:2.2;
```

```
    transition:.3s;
```

```
}
```

```
footer a:hover{
```

```
    color:white;
```

```
}
```

```
.footer-small{
```

```
    color:var(--primary);

    font-weight:600;
}

.footer-bottom{

    color:#6B7280;

    text-align:center;

    margin-top:25px;

    font-size:14px;
}

/* =====
SCROLLBAR
===== */

::-webkit-scrollbar{

    width:8px;
}

::-webkit-scrollbar-track{

    background:#0D1117;
}

::-webkit-scrollbar-thumb{

    background:#374151;

    border-radius:20px;
}

::-webkit-scrollbar-thumb:hover{

    background:var(--primary);
```

```
}
```

```
/* =====  
RESPONSIVE  
===== */
```

```
@media(max-width:992px){
```

```
.hero{
```

```
    text-align:center;
```

```
    min-height:auto;
```

```
    padding-top:140px;
```

```
}
```

```
.hero-image{
```

```
    justify-content:center;
```

```
    margin-top:60px;
```

```
}
```

```
.hero h1{
```

```
    font-size:54px;
```

```
}
```

```
.hero h2{
```

```
    font-size:24px;
```

```
}
```

```
.feature-grid{
```

```
    grid-template-columns:1fr;
```

```
}
```

```
.hero-buttons{

    justify-content:center;

}

}

@media(max-width:768px){

.hero h1{

    font-size:42px;

}

.hero p{

    font-size:16px;

}

.section-title h2{

    font-size:34px;

}

.cta h2{

    font-size:32px;

}

}

/* =====
Smooth Reveal
===== */

.feature-card,
.workflow-step,
.cta,
.hero-image,
.section-title{
```

```

    opacity:0;

    transform:translateY(40px);

}

.hero-image i{

    transition:.4s;

}

/* =====
VIVARX BADGE
===== */

.project-badge{

    position:relative;

    display:inline-flex;

    align-items:center;

    gap:10px;

    padding:8px 18px;

    background:#161B22;

    border:1px solid #30363D;

    border-radius:30px;

    overflow:hidden;

}

/* Green Live Dot */

.project-badge::before{

    content:"";

    width:8px;

```

```

height:8px;

border-radius:50%;

background:#22C55E;

display:inline-block;

animation:liveDot 2s infinite;

}

@keyframes liveDot{

  0%,100%{

    opacity:.4;

    box-shadow:0 0 0 rgba(34,197,94,0);

  }

  50%{

    opacity:1;

    box-shadow:0 0 10px rgba(34,197,94,.8);

  }

}

/* Animated VivarX */

.project-name{

  background:linear-gradient(
    90deg,
    #E11D48,
    #FF6A95,
    #E11D48
  );

  background-size:200% auto;

```



```
-webkit-background-clip:text;

-webkit-text-fill-color:transparent;

animation:vivarxGradient 4s linear infinite;

font-weight:700;

}

@keyframes vivarxGradient{

  from{

    background-position:0%;

  }

  to{

    background-position:200%;

  }

}

/* =====
POWERED BY VIVARX
===== */

.project-badge{

  display:inline-flex;
  align-items:center;
  gap:10px;

  background:#161B22;
  border:1px solid #30363D;
  border-radius:30px;

  padding:10px 18px;

  transition:.35s ease;
```

```
}
```

```
.project-badge::before{
```

```
  content:"";
```

```
  width:8px;
```

```
  height:8px;
```

```
  border-radius:50%;
```

```
  background:#22C55E;
```

```
  animation:statusPulse 2s infinite;
```

```
}
```

```
.project-badge:hover{
```

```
  border-color:#E11D48;
```

```
  transform:translateY(-2px);
```

```
}
```

```
.project-text{
```

```
  color:#9AA3B2;
```

```
  font-size:14px;
```

```
  font-weight:500;
```

```
}
```

```
.project-name{
```

```
  color:#E11D48;
```

```
  font-weight:700;
```

```
  position:relative;
```

```
  overflow:hidden;
```

```
}
```

```
.project-name::after{
```

```
    content:"";
```

```
    position:absolute;
```

```
    top:0;
```

```
    left:-120%;
```

```
    width:40%;
```

```
    height:100%;
```

```
    background:linear-gradient(
```

```
        90deg,
```

```
        transparent,
```

```
        rgba(255,255,255,.9),
```

```
        transparent
```

```
    );
```

```
    transform:skewX(-20deg);
```

```
    animation:scanEffect 3s infinite;
```

```
}
```

```
@keyframes scanEffect{
```

```
    0%{
```

```
        left:-120%;
```

```
    }
```

```
    100%{
```

```
        left:140%;
```

```
    }
```

```
}
```

```
@keyframes statusPulse{
```

```
    0%,100%{
```

```
        opacity:.4;
```

```
        box-shadow:0 0 0 rgba(34,197,94,0);
```

```

    }

    50%{

        opacity:1;
        box-shadow:0 0 10px rgba(34,197,94,.8);

    }

}

/* =====
    UPLOAD PAGE
===== */

.upload-page{

    padding-top:140px;
    padding-bottom:100px;

}

.upload-card{

    background:#161B22;
    border:1px solid #30363D;
    border-radius:22px;
    padding:60px;
    text-align:center;
    transition:.35s;

}

.upload-card:hover{

    border-color:#E11D48;
    box-shadow:0 20px 45px rgba(225,29,72,.10);

}

.upload-icon{

    width:90px;
    height:90px;

```

```
border-radius:50%;

margin:auto;

display:flex;
align-items:center;
justify-content:center;

background:#1F2937;

color:#E11D48;

font-size:42px;

margin-bottom:30px;
}
```

```
.upload-card h3{

font-weight:700;
margin-bottom:10px;
}
```

```
.upload-card p{

color:#9AA3B2;
}
```

```
.selected-file{

margin-top:25px;

color:#CBD5E1;

font-weight:600;
}
```

```
.supported-files{

margin-top:35px;
```

```
}
```

```
.supported-files span:first-child{
```

```
    color:#9AA3B2;
```

```
    font-size:14px;
```

```
}
```

```
.file-pill{
```

```
    display:inline-block;
```

```
    padding:8px 15px;
```

```
    margin:6px;
```

```
    border-radius:30px;
```

```
    background:#1C2128;
```

```
    border:1px solid #30363D;
```

```
    color:#CBD5E1;
```

```
    font-size:13px;
```

```
}
```

```
.file-pill:hover{
```

```
    border-color:#E11D48;
```

```
    color:white;
```

```
}
```

```
/* =====
```

```
SECURITY CARDS
```

```
===== */
```

```
.security-box{
```

```
background:#161B22;

border:1px solid #30363D;

border-radius:18px;

padding:35px;

height:100%;

transition:.35s;

}
```

```
.security-box:hover{

    transform:translateY(-8px);

    border-color:#E11D48;

}
```

```
.security-box i{

    font-size:40px;

    color:#E11D48;

    margin-bottom:20px;

}
```

```
.security-box h5{

    font-weight:700;

    margin-bottom:15px;

}
```

```
.security-box p{

    color:#9AA3B2;

    line-height:1.8;
```

```
}
```

```
/* =====  
DRAG AREA (Future)  
===== */
```

```
.drop-zone{  
  
    border:2px dashed #30363D;  
  
    border-radius:18px;  
  
    padding:60px;  
  
    transition:.35s;  
  
}
```

```
.drop-zone:hover{  
  
    border-color:#E11D48;  
  
}
```

```
/* =====  
ANALYZE BUTTON  
===== */
```

```
.upload-card .btn-danger{  
  
    min-width:220px;  
  
    font-weight:600;  
  
    letter-spacing:.5px;  
  
}
```

```
/* =====  
RESPONSIVE  
===== */
```

```
@media(max-width:768px){
```



```
.upload-card{  
  
    padding:35px 25px;  
  
}
```

```
.upload-icon{  
  
    width:70px;  
    height:70px;  
    font-size:34px;  
  
}
```

```
.file-pill{  
  
    margin:4px;  
  
}
```

```
.recent-analysis{  
  
    overflow-x:auto;  
  
}
```

```
}  
/* =====  
    ANALYZE BUTTON  
===== */
```

```
.analyze-btn{  
  
    position:relative;  
  
    overflow:hidden;  
  
    min-width:260px;  
  
    height:58px;  
  
    font-weight:600;  
  
    letter-spacing:.5px;
```

```
display:inline-flex;

align-items:center;

justify-content:center;

gap:12px;

}

.analyze-btn i{

    font-size:22px;

}

.analyze-btn::before{

    content:"";

    position:absolute;

    top:0;

    left:-120%;

    width:45%;

    height:100%;

    background:linear-gradient(

        90deg,

        transparent,

        rgba(255,255,255,.45),

        transparent

    );

    transform:skewX(-25deg);

    animation:buttonScan 3s linear infinite;
```

```
}
```

```
@keyframes buttonScan{
```

```
  100%{
```

```
    left:160%;
```

```
  }
```

```
}
```

```
.analyze-btn:hover{
```

```
  transform:translateY(-3px);
```

```
  box-shadow:0 15px 35px rgba(225,29,72,.25);
```

```
}
```

```
.upload-icon{
```

```
  animation:uploadPulse 3s ease-in-out infinite;
```

```
}
```

```
@keyframes uploadPulse{
```

```
  0%,100%{
```

```
    transform:scale(1);
```

```
    box-shadow:0 0 0 rgba(225,29,72,0);
```

```
  }
```

```
  50%{
```

```
    transform:scale(1.08);
```

```
    box-shadow:0 0 20px rgba(225,29,72,.35);
```

```
  }
```

```

}
/* =====
    INVESTIGATION REPORT
===== */

.dashboard-page{

    padding-top:140px;
    padding-bottom:120px;

}

/* =====
    VERDICT CARD
===== */

.verdict-card{

    background:linear-gradient(
        135deg,
        #2A1016,
        #161B22
    );

    border:1px solid #E11D48;

    border-radius:20px;

    padding:45px;

    box-shadow:0 20px 45px rgba(225,29,72,.15);

}

.verdict-tag{

    display:inline-block;

    padding:8px 18px;

    background:#E11D48;

    color:white;

    border-radius:30px;

```

```
font-size:13px;

font-weight:600;

letter-spacing:1px;

}
```

```
.verdict-card h2{

font-weight:700;

margin-top:15px;

}
```

```
.verdict-card p{

color:#CBD5E1;

margin-top:12px;

}
```

```
/* =====
RISK SCORE
===== */
```

```
.risk-circle{

width:140px;

height:140px;

margin-left:auto;

border-radius:50%;

border:8px solid #E11D48;

display:flex;

align-items:center;
```

```
justify-content:center;

font-size:34px;

font-weight:700;

color:white;

background:#1C2128;

box-shadow:0 0 35px rgba(225,29,72,.30);

}
```

```
.risk-label{

display:block;

margin-top:15px;

color:#9AA3B2;

font-size:15px;

}
```

```
/* =====
REPORT CARD
===== */
```

```
.report-card{

background:#161B22;

border:1px solid #30363D;

border-radius:18px;

padding:30px;

height:100%;

transition:.35s;

}
```

```

.report-card:hover{

    border-color:#E11D48;

    transform:translateY(-5px);

    box-shadow:0 18px 40px rgba(225,29,72,.12);

}

.report-card h4{

    margin-bottom:25px;

    font-weight:700;

}

/* =====
TABLE
===== */

.report-card table{

    margin-bottom:0;

}

.report-card td{

    color:#CBD5E1;

    border-color:#30363D;

}

.report-card th{

    color:white;

    border-color:#30363D;

}

/* =====

```

COPY BUTTON

===== */

```
.copy-btn{
  border:none;
  background:#1C2128;
  color:#CBD5E1;
  border:1px solid #30363D;
  border-radius:8px;
  padding:6px 14px;
  font-size:13px;
  transition:.3s;
}
```

```
.copy-btn:hover{
  background:#E11D48;
  border-color:#E11D48;
  color:white;
}
```

/* =====

ENTROPY

===== */

```
.entropy-item{
  margin-bottom:20px;
}
```

```
.entropy-item span{
  display:block;
  margin-bottom:8px;
  font-weight:600;
  color:white;
}
```

```
.entropy-item small{
  display:block;
  margin-top:8px;
  color:#9AA3B2;
}
```

```
.progress{
  height:10px;
  background:#242B35;
  border-radius:30px;
```



```

        overflow:hidden;
    }

    .progress-bar{
        border-radius:30px;
    }

    /* =====
    STRINGS
    ===== */

    .strings-box{
        display:flex;
        flex-wrap:wrap;
        gap:12px;
    }

    .string-pill{
        padding:10px 18px;
        border-radius:30px;
        background:#1C2128;
        border:1px solid #30363D;
        color:#CBD5E1;
        font-family:Consolas, monospace;
        transition:.3s;
    }

    .string-pill:hover{
        border-color:#E11D48;
        background:#23141A;
        color:white;
        transform:translateY(-2px);
    }

    /* =====
    DOWNLOAD BUTTON
    ===== */

    .download-btn{
        min-width:260px;
        height:58px;
        border:none;
        border-radius:10px;
        background:#E11D48;
        color:white;
    }

```

```
font-weight:600;
display:inline-flex;
align-items:center;
justify-content:center;
gap:12px;
transition:.35s;
}
```

```
.download-btn:hover{
  background:#F43F5E;
  transform:translateY(-3px);
  box-shadow:0 18px 35px rgba(225,29,72,.25);
}
```

```
/* =====
CARD ANIMATION
===== */
```

```
.report-card{
  animation:reportFade .7s ease;
}
```

```
@keyframes reportFade{
  from{
    opacity:0;
    transform:translateY(25px);
  }
  to{
    opacity:1;
    transform:translateY(0);
  }
}
```

```
/* =====
ERROR BOX
===== */
```

```
.error-box{
  background:#23141A;
  border:1px solid #E11D48;
  border-radius:10px;
  padding:18px;
  color:#F4B4C0;
}
```

```
/* =====  
MOBILE  
===== */
```

```
@media(max-width:992px){  
  .risk-circle{  
    margin:30px auto 0;  
  }  
  .verdict-card{  
    text-align:center;  
  }  
}
```

```
@media(max-width:768px){  
  .dashboard-page{  
    padding-top:110px;  
  }  
  .verdict-card{  
    padding:30px;  
  }  
  .risk-circle{  
    width:110px;  
    height:110px;  
    font-size:26px;  
  }  
  .report-card{  
    padding:22px;  
  }  
  .string-pill{  
    width:100%;  
    text-align:center;  
  }  
  .download-btn{  
    width:100%;  
  }  
}
```

```
/* =====  
VERDICT SEVERITY VARIANTS  
===== */
```

```
.verdict-tag.risk-high{ background:#E11D48; }  
.verdict-tag.risk-medium{ background:#F59E0B; color:#1C2128; }  
.verdict-tag.risk-low{ background:#3B82F6; }  
.verdict-tag.risk-clean{ background:#22C55E; color:#0D1117; }
```

```
.risk-circle.risk-high{ border-color:#E11D48; color:#E11D48; }
.risk-circle.risk-medium{ border-color:#F59E0B; color:#F59E0B; }
.risk-circle.risk-low{ border-color:#3B82F6; color:#3B82F6; }
.risk-circle.risk-clean{ border-color:#22C55E; color:#22C55E; }
```

```
/* =====
  GENERIC INNER PAGE HERO
  (Features / Documentation / Contact)
  ===== */
```

```
.page-hero{
  padding-top:160px;
  padding-bottom:60px;
}
```

```
.page-hero h1{
  font-size:46px;
  font-weight:800;
}
```

```
.features-section.pt-0{
  padding-top:20px;
}
```

```
@media(max-width:768px){
  .page-hero{
    padding-top:130px;
    padding-bottom:30px;
  }
  .page-hero h1{
    font-size:32px;
  }
}
```

```
/* =====
  DOCUMENTATION PAGE
  ===== */
```

```
.docs-section{
  padding-bottom:100px;
}
```

```
.doc-toc{
  position:sticky;
```

```
top:110px;
display:flex;
flex-direction:column;
gap:10px;
border-left:1px solid var(--border);
padding-left:20px;
}
```

```
.doc-toc h6{
  color:var(--muted);
  text-transform:uppercase;
  letter-spacing:1px;
  font-size:12px;
  margin-bottom:6px;
}
```

```
.doc-toc a{
  color:var(--muted);
  text-decoration:none;
  font-size:14px;
  transition:.25s;
}
```

```
.doc-toc a:hover{
  color:white;
  padding-left:4px;
}
```

```
.doc-block{
  margin-bottom:55px;
}
```

```
.doc-block h3{
  font-size:26px;
  font-weight:700;
  margin-bottom:18px;
  padding-bottom:14px;
  border-bottom:1px solid var(--border);
}
```

```
.doc-block h5{
  font-size:17px;
  font-weight:600;
  margin:24px 0 12px;
}
```

```
.doc-block p{
  color:var(--muted);
  line-height:1.9;
  margin-bottom:14px;
}
```

```
.doc-block code{
  background:var(--card);
  border:1px solid var(--border);
  border-radius:6px;
  padding:2px 8px;
  color:var(--primary);
  font-size:.9em;
}
```

```
.doc-block a{
  color:var(--primary);
}
```

```
.doc-list{
  color:var(--muted);
  line-height:1.9;
  padding-left:22px;
  margin-bottom:14px;
}
```

```
.doc-list li{
  margin-bottom:6px;
}
```

```
.doc-code{
  background:var(--card);
  border:1px solid var(--border);
  border-radius:12px;
  padding:22px 24px;
  overflow-x:auto;
  margin-bottom:18px;
}
```

```
.doc-code code{
  background:none;
  border:none;
  padding:0;
  color:#CBD5E1;
```

```
font-family:Consolas, monospace;
font-size:13.5px;
line-height:1.7;
white-space:pre;
}

/* =====
CONTACT CARDS
===== */

.contact-card{
  display:block;
  background:var(--surface);
  border:1px solid var(--border);
  border-radius:18px;
  padding:40px;
  text-align:center;
  text-decoration:none;
  color:var(--text);
  height:100%;
  transition:.35s ease;
}

.contact-card:hover{
  transform:translateY(-8px);
  border-color:var(--primary);
  box-shadow:0 18px 40px rgba(225,29,72,.12);
  color:white;
}

.contact-card i{
  font-size:38px;
  color:var(--primary);
  margin-bottom:18px;
  display:block;
}

.contact-card h4{
  font-weight:700;
  margin-bottom:8px;
}

.contact-card p{
  color:var(--muted);
  margin:0;
```

```

    word-break:break-word;
}

/* =====
NAV ACTIVE STATE
===== */

.nav-link.active{
    position:relative;
}

.nav-link.active::after{
    content:"";
    position:absolute;
    left:0;
    right:0;
    bottom:-6px;
    height:2px;
    background:var(--primary);
    border-radius:2px;
}

/* =====
DOCS RESPONSIVE
===== */

@media(max-width:992px){
    .doc-toc{
        position:static;
        flex-direction:row;
        flex-wrap:wrap;
        border-left:none;
        border-bottom:1px solid var(--border);
        padding-left:0;
        padding-bottom:20px;
        margin-bottom:30px;
        gap:14px 22px;
    }
}

@media(max-width:768px){
    .doc-block h3{
        font-size:21px;
    }
    .doc-code{

```



```

        padding:16px;
    }
    .doc-code code{
        font-size:12px;
    }
    .contact-card{
        padding:28px;
    }
}

/* =====
PAGE LOAD MOTION
===== */

body{
    animation:pageFadeIn .5s ease;
}

@keyframes pageFadeIn{
    from{ opacity:0; }
    to{ opacity:1; }
}

@media(prefers-reduced-motion: reduce){
    *, *::before, *::after{
        animation-duration:.01ms !important;
        animation-iteration-count:1 !important;
        transition-duration:.01ms !important;
        scroll-behavior:auto !important;
    }
}

/* =====
MOBILE NAV
===== */

@media(max-width:991px){
    .navbar-collapse{
        background:rgba(13,17,23,.98);
        margin-top:14px;
        padding:18px;
        border-radius:14px;
        border:1px solid var(--border);
    }
    .navbar-nav{

```

```

        align-items:flex-start !important;
        gap:6px;
    }
    .nav-link{
        margin-left:0 !important;
        padding:8px 0;
    }
    .nav-link.active::after{
        display:none;
    }
    .navbar-nav .nav-item.ms-3{
        margin-left:0 !important;
        margin-top:10px;
        width:100%;
    }
    .navbar-nav .nav-item.ms-3 .btn{
        width:100%;
    }
}

```

static/js/main.js

```

/* =====
MallnSight UI
===== */

document.addEventListener("DOMContentLoaded", () => {

    /* =====
    Navbar Scroll Effect
    ===== */

    const navbar = document.querySelector(".navbar");

    window.addEventListener("scroll", () => {

        if (window.scrollY > 40) {

            navbar.style.background = "rgba(13,17,23,.97)";
            navbar.style.boxShadow = "0 8px 25px rgba(0,0,0,.35)";

        } else {

            navbar.style.background = "rgba(13,17,23,.90)";
            navbar.style.boxShadow = "none";

```

```

    }

  });

  /* =====
  Reveal Animation
  ===== */

  const reveals = document.querySelectorAll(
    ".feature-card,.workflow-step,.cta,.hero-image,.section-title," +
    ".doc-block,.contact-card"
  );

  const observer = new IntersectionObserver((entries)=>{

    entries.forEach(entry=>{

      if(entry.isIntersecting){

        entry.target.style.opacity="1";
        entry.target.style.transform="translateY(0px)";

        observer.unobserve(entry.target);

      }

    });

  },{

    threshold:0.15

  });

  reveals.forEach((item,index)=>{

    item.style.opacity="0";
    item.style.transform="translateY(40px)";
    item.style.transition=`opacity .6s cubic-bezier(.16,1,.3,1) ${index%3}*0.08}s,
      transform .6s cubic-bezier(.16,1,.3,1) ${index%3}*0.08}s`;

    observer.observe(item);

  });

```

```

/* =====
Card Hover Tilt
===== */

document.querySelectorAll(".feature-card").forEach(card=>{

  card.addEventListener("mousemove",(e)=>{

    const rect = card.getBoundingClientRect();

    const x = e.clientX - rect.left;
    const y = e.clientY - rect.top;

    const rotateX = (y - rect.height/2)/25;
    const rotateY = -(x - rect.width/2)/25;

    card.style.transform=
      `perspective(800px)
      rotateX(${rotateX}deg)
      rotateY(${rotateY}deg)
      translateY(-6px)`;

  });

  card.addEventListener("mouseleave",()=>{

    card.style.transform="translateY(0px)";

  });

});

/* =====
Hero Icon Pulse
===== */

const heroIcon = document.querySelector(".hero-image i");

if(heroIcon){

  setInterval(()=>{

    heroIcon.animate([

```

```
        {
            transform:"scale(1)"
        },

        {
            transform:"scale(1.08)"
        },

        {
            transform:"scale(1)"
        }

    ],{

        duration:1800

    });

    },3500);

}

});
```

Tests

tests/test_app.py

```
import io
import os
import sys

sys.path.insert(0, os.path.abspath(os.path.join(os.path.dirname(__file__), "..")))

import pytest

from app import app

@pytest.fixture
def client():
    app.config["TESTING"] = True
    with app.test_client() as client:
        yield client

@pytest.mark.parametrize("route", ["/", "/about", "/features", "/upload", "/contact", "/history"])
def test_static_pages_load(client, route):
    response = client.get(route)
    assert response.status_code == 200

def test_analyze_requires_a_file(client):
    response = client.post("/analyze", data={})
    assert response.status_code == 400

def test_analyze_rejects_get(client):
    response = client.get("/analyze")
    assert response.status_code == 405

def test_analyze_rejects_disallowed_extension(client):
    data = {
        "file": (io.BytesIO(b"hello world"), "notes.txt")
    }
    response = client.post("/analyze", data=data, content_type="multipart/form-data")
    assert response.status_code == 400
```

```

def test_history_degrades_gracefully_without_mongodb_uri(client, monkeypatch):
    from analysis import history

    monkeypatch.delenv("MONGODB_URI", raising=False)
    history._reset_cache()

    response = client.get("/history")

    assert response.status_code == 200
    assert b"unavailable" in response.data.lower()

    history._reset_cache()


def test_analyze_accepts_pe_file(client):
    pe_path = os.path.join(os.path.dirname(sys.executable), "python.exe")

    if not os.path.exists(pe_path):
        pytest.skip("No PE binary available to test with on this platform")

    with open(pe_path, "rb") as f:
        data = {
            "file": (io.BytesIO(f.read()), "sample.exe")
        }
        response = client.post("/analyze", data=data, content_type="multipart/form-data")

    assert response.status_code == 200
    assert b"Risk Score" in response.data

```

Configuration & Dependencies

requirements.txt

```
blinker==1.9.0
click==8.4.2
Flask==3.1.3
itsdangerous==2.2.0
Jinja2==3.1.6
MarkupSafe==3.0.3
pefile==2024.8.26
pymongo==4.17.0
python-dotenv==1.2.2
reportlab==5.0.0
waitress==3.0.2
Werkzeug==3.1.8
yara-python==4.5.4
```

.env.example

```
# Copy this file to .env and fill in your own values.
# .env is gitignored — never commit real credentials.

# MongoDB Atlas connection string (used for the /history cloud feature).
# Get this from your Atlas cluster: Database > Connect > Drivers.
# Example: mongodb+srv://<user>:<password>@<cluster-host>/?retryWrites=true&w=majority
MONGODB_URI=

# Optional: database name (defaults to "mallnsight" if not set)
MONGODB_DB=mallnsight

# Optional: enable Flask debug mode locally (never enable in production)
FLASK_DEBUG=0
```

Procfile

```
web: waitress-serve --host=0.0.0.0 --port=$PORT app:app
```

runtime.txt

```
python-3.12.7
```